Reproducible Science

Methodological School - 3 R’s of Trustworthy Science

Margherita Calderan

May 18, 2026

What is reproducible science?

Reproducibility can be considered as the most fundamental pre-requisite of replication in science.

> "…obtaining consistent results using the same input data, computational steps, methods, and code, and conditions of analysis." Reproducibility et al. (2019)

Meaning that someone else, or even you, in the future, can **reproduce** your **results** from your **materials**: your data, your code, your documentation.

Our job is hard

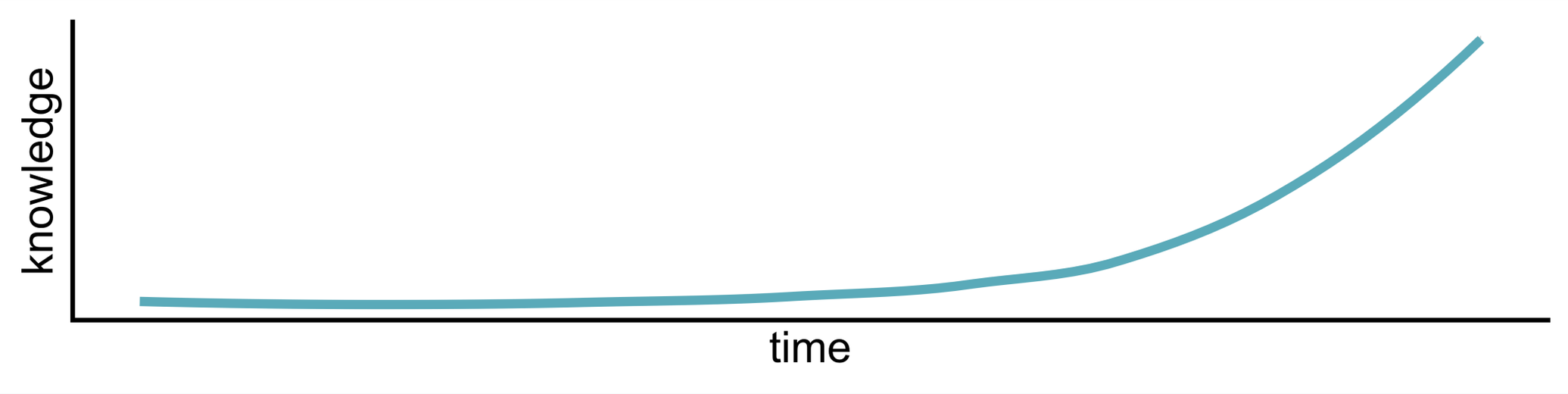
- Running experiments
- Analyzing data
- Managing trainees
- Writing papers
- Responding to reviewers

Reproducibility helps!

- **Organizes** your workflow.
- **Saves time** by documenting steps.
- **Builds trust** in your findings.
- Enables others to **reproduce** and **extend** your work.

Keys to reproducible science

- **Data**: organize, document, and share your datasets.
- **Code**: write analysis scripts that are clean, transparent, and reusable.
- **Literate programming**: combine code and text in the same document, so your reports are dynamic and replicable.
- **Version Control and Sharing**: track changes, collaborate, and make your work openly available using tools like GitHub and OSF (and/or Zenodo).



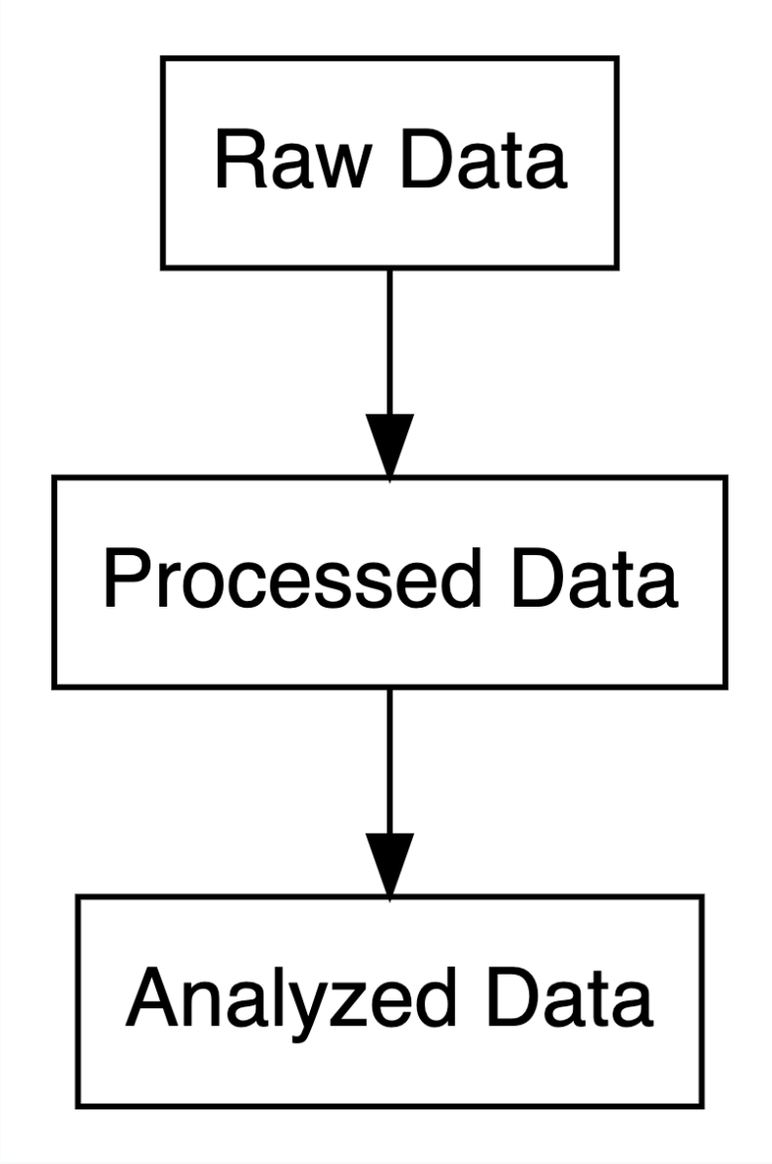
Outline

1. Data
2. Code
3. R projects
4. Literate Programming
5. Version Control

Data

Data types in research

- **Raw Data**: Original, unprocessed (e.g., survey responses).
- **Processed Data**: Cleaned, digitized, or compressed.
- **Analyzed Data**: Summarized in tables, charts, or text.



Share all your data: where?

- dryad
- zenodo
- osf

Why use dryad?

- **Flexible**: Accepts any file format from any field.
- **Citable**: Provides a persistent DOI.
- **Curated**: Data curators check files for best practices.
- **Integrated**: Streamlines sharing with publishers (Wiley, Royal Society, PLOS).

Why use zenodo?

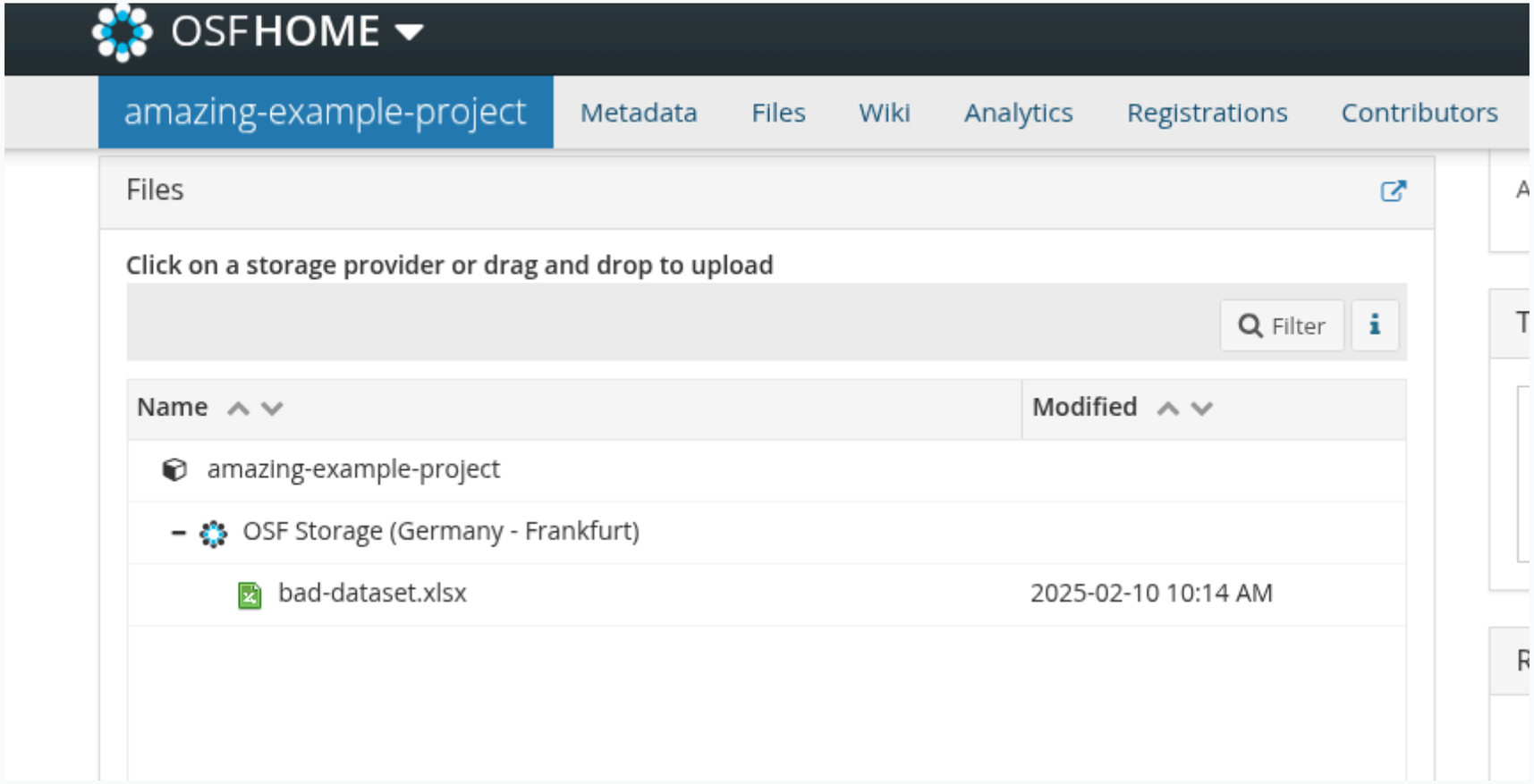
- **Secure**: Stored safely in CERN's Data Centre.
- **Citable**: Every upload gets a DOI.
- **Controlled Access**: Can restrict access for sensitive data (e.g., clinical trials).
- **GitHub Integration**: Easily preserve your GitHub repositories.

Why use osf?

- **Collaborative**: Add collaborators and manage projects
- **Citable**: Every file gets a unique URL for citing, project gets a DOI.
- **Comprehensive**: Automate version control, preregister research, and share preprints.
- **Long-term**: Guaranteed 50+ years of read access hosting
- **GitHub Integration**: Easily preserve your GitHub repositories.

Bad data sharing example

Imagine this scenario: you read a paper that seems really relevant to your research. At the end, you're excited to see they've shared their data on OSF. You go to the repository, and there's one file…



Bad data sharing example

You download it, open it, and you see this…

x1	x2	x3	x4	x5	x6	x7
0.3981105	13.912435	a	0	-0.6775811	0.8759740	-0.2051604
-0.1434733	1.093743	c	0	0.7055193	0.2521987	1.8816947
-0.2526000	4.898035	c	0	0.4744651	-0.5628840	0.3245589
-1.2272588	14.717053	b	0	-0.5132792	-1.1368242	-0.1355150
-0.4360417	8.547025	c	1	-0.1736804	-0.7120962	-1.2714320

What do these variables mean? What's x3?

What do 0 and 1 represent?

How are missing values coded?

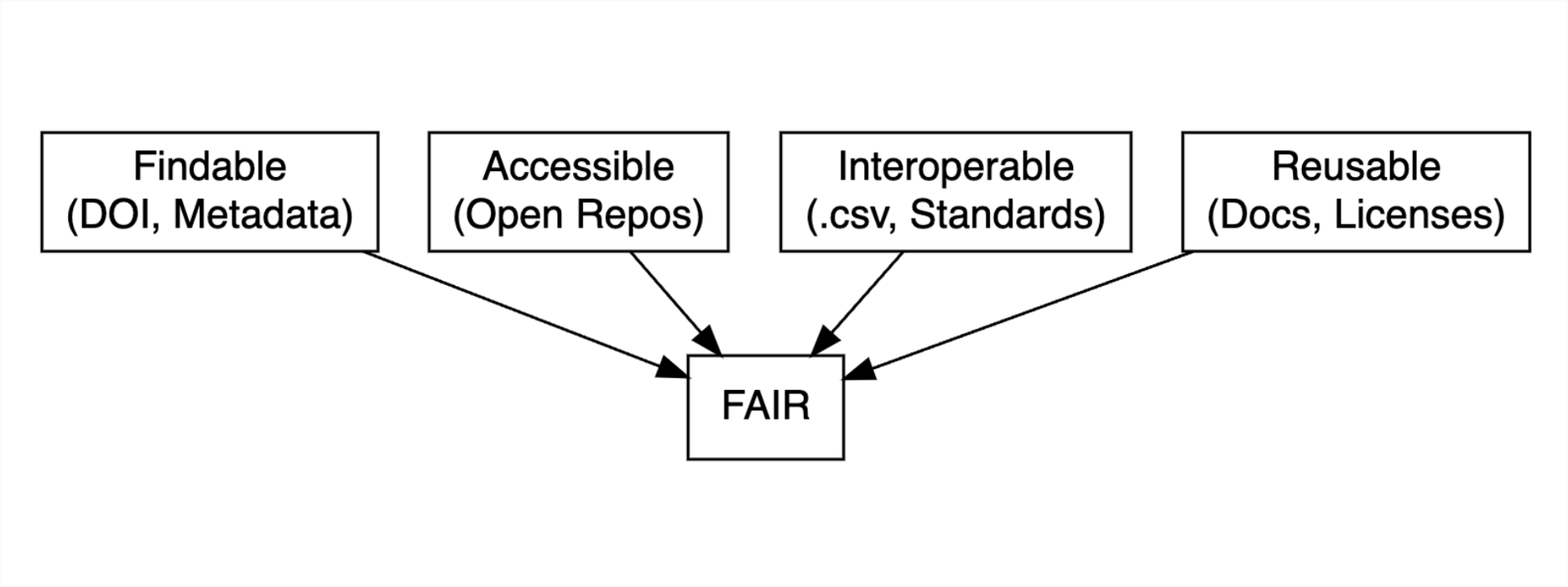
Is x6 a z-score or raw data?

Good data sharing practices

- Use **plain-text formats** (e.g., `.csv`, `.txt`).
- Include a **data dictionary** with variable descriptions.
- Add a **README** with key details.
- Follow **FAIR principles** (Findable, Accessible, Interoperable, Reusable; Wilkinson et al. (2016), https://www.go-fair.org/fair-principles/)

FAIR data principles

- **Findable**: Use metadata and DOIs to make data easy to locate.
- **Accessible**: Ensure data is retrievable via open repositories.
- **Interoperable**: Use standard formats (e.g., `.csv`, `.txt`) for compatibility.
- **Reusable**: Include clear documentation and open licenses.

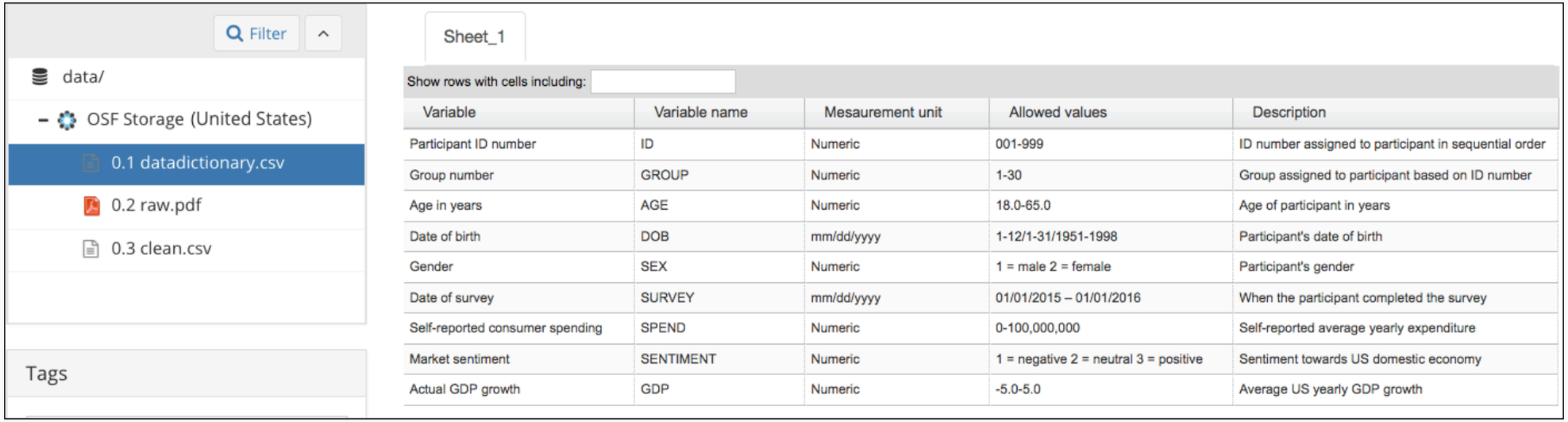


Data dictionary

A data dictionary is a document that outlines the structure, content, and variable definitions for a dataset (harvard/datamanagement).

It is critical for reproducibility because it explains what all the variable names and values in your spreadsheet really mean (osf/datadictionary).

Data dictionary



Variable	Variable name	Mesaurement unit	Allowed values	Description
Participant ID number	ID	Numeric	001-999	ID number assigned to participant in sequential order
Group number	GROUP	Numeric	1-30	Group assigned to participant based on ID number
Age in years	AGE	Numeric	18.0-65.0	Age of participant in years
Date of birth	DOB	mm/dd/yyyy	1-12/1-31/1951-1998	Participant's date of birth
Gender	SEX	Numeric	1 = male 2 = female	Participant's gender
Date of survey	SURVEY	mm/dd/yyyy	01/01/2015 – 01/01/2016	When the participant completed the survey
Self-reported consumer spending	SPEND	Numeric	0-100,000,000	Self-reported average yearly expenditure
Market sentiment	SENTIMENT	Numeric	1 = negative 2 = neutral 3 = positive	Sentiment towards US domestic economy
Actual GDP growth	GDP	Numeric	-5.0-5.0	Average US yearly GDP growth

From OSF guide

- Variable names
- Human-readable variable names
- Measurement units for the variable
- Allowed values for the variable
- Definition of the variable

Data dictionary: let's try

Imagine we are collecting data (n = 12) to explore the relationship between anxiety (measured via the State-**Trait** Anxiety Inventory) and education levels:

▶ Code

```
   id anxi edu
1  a6   50 PhD
2  b5   59 PhD
3  c9   37 PhD
4  d3   46 PhD
5  e7   48 BSc
6  f6   44 BSc
7  g7   56 BSc
8  h4   54 BSc
9  i6   42 MSc
10 j4   48 MSc
11 k6   48 MSc
12 l4   37 MSc
```

> STAI-T consists of 20 items, 4-points likert scale

datadictionary

```
library(datadictionary)

descr <- list(
  anxi = "Anxiety symptoms measured with the State-Trait Anxiety Inventory
  edu  = "Highest degree obtained, factor with three levels: PhD, BSc, MSc'

datdi <- create_dictionary(df, id_var = "id",
                           var_labels = descr)
```

Data dictionary

▶ Code

item	label	class	summary	value
			Rows in dataset	12
			Columns in dataset	3
id	Unique identifier		unique values	12
			missing	0
anxi	Anxiety symptoms measured with the State-Trait Anxiety Inventory (state scale); 20 items summed on a 4-point Likert scale.	integer	mean	47
			median	48
			min	37
			max	59
			missing	0
edu	Highest degree obtained, factor with three levels: PhD, BSc, MSc	factor	BSc (1)	4
			MSc (2)	4
			PhD (3)	4
			missing	0

Data dictionary - Good data sharing example

The daily costs of workaholism

Data-dictionary.txt

```
- `ID`: Factor indexing participants identification codes (e.g., "S001", "S002", etc.)

- `day`: Integer indexing the day of participation (from 1 to 10)

- `SBP_aft`, `DBP_aft`: Numeric indexing afternoon measurements (mmHg) of systolic and diastolic blood pressure, respectively. Note: these
variables have been computed by averaging each pair of consecutive recordings

- `WHLSM1` ... `WHLSM6`: Item scores at the state workaholism measure (1–7) administered in the afternoon

- `SBP_eve`, `DBP_eve`: Numeric indexing evening measurements (mmHg) of systolic and diastolic blood pressure, respectively. Note: these
variables have been computed by averaging each pair of consecutive recordings

- `EE1` ... `EE4`: Item scores at the emotional exhaustion measure (1–7) administered in the evening

- `R.det1` ... `R.det3`: Item scores at the psychological detachment measure (1–7) administered in the evening

- `SQ1` ... `SQ4`: Item scores at the sleep disturbances measure (1–7) administered in the morning. Note: higher values indicate worse
sleep disturbances; these measures are related to the following day

- `gender`: Factor indexing participants' gender ("F" or "M")

- `age`: Integer indexing participants' age (years)

- `BMI`: Numeric indexing participants' body mass index ($kg/m^2$)

- `IN.resp`: Factor indexing whether the participant was included in the main sample based on the response rate inclusion criteria (TRUE)
or not (FALSE)

- `IN.bp`: Factor indexing whether the participant was included in the sample considered for blood pressure analyses (TRUE) or not (FALSE)
```

README

A **README** file is the first thing someone sees when they open your dataset (or project folder). It should answer basic questions like:

- What is this dataset?
- How was it collected?
- What are the variables?
- Which is the structure of the project?

README

Anxiety and Education Example Dataset

This repository contains a small simulated dataset with 12 observations, designed to explore the relationship between anxiety and education level.

Anxiety is measured using the State-Trait Anxiety Inventory (STAI), specifically the trait scale. Education level is recorded as the highest degree obtained (Bachelor, Master, PhD).

README - Good example

Wearing face masks when no longer mandatory: An exploratory study about a...

README.txt

```
This repository "data" contains a CSV file that serves as the data source for the network analysis and regression analysis discussed in
the accompanying article. Each column within the CSV file corresponds to a node in the network analysis.

Run first the Network_analysis.R script and then the "Generalized linear mixed effect modeling.R script.
```

README - Good example

README.md

Tracking the Unconscious: Neural evidence for the retention of unaware information in visual working memory

This repository contains the code to reproduce the analysis, figures and tables of the paper *Tracking the Unconscious: Neural evidence for the retention of unaware information in visual working memory* by Gambarota et al. The repository contains also the code to run the experiment using Psychopy along with raw EEG and behavioral data.

Structure

- The `data` folder contains the raw and cleaned EEG and behavioral data:

<!-- -->

```
data
├── clean
│   ├── behavioral
│   └── eeg
└── raw
    ├── behavioral
    │   ├── csv
    │   └── pkl
    └── eeg
```

- The `experiment/` folder contains python scripts to run the experiment.

<!-- -->

```
experiment
├── experiment.py
└── triggers.py
```

Data licensing

A license tells others what they can and can’t do with your data. If you don’t include one, legally speaking, people might not be allowed to use it, even if you meant to share it openly.

Data licensing

Common licenses for documents, data, and other non-software:

- CC BY: Permits reuse of documents, data, and other non-software works with attribution.
- CC0: Places documents, data, or other works in the public domain; no restrictions on reuse.

Data licensing

Common licenses for software:

- GNU GPL-3.0: Applies to software and guarantees freedom to run, study, share, and modify software, requiring modified versions be distributed under the same license.
- AGPL-3.0: Extends the GNU GPL by requiring source code to be made available if a modified version runs on a publicly accessible server.

Code

What are the alternatives?

There are several excellent open-source software options based on R, such as:

- Jamovi
  - **Pros**: access to the underlying R code
  - **Cons**: still limited in functions, graphics, and complex models
- JASP
  - **Pros**: many models, including advanced ones
  - **Cons**: no access to the underlying R code

Point and click workflow

- Click menu items to run analysis
- “exclude <18”
- Click through everything again
- Forget a step? Round differently?

> Stressful, error-prone, and undocumented.

Why scripting?

R workflow:

```r
# Load data
data <- read.csv("data.csv")

# Filter age
data <- data[data$age >= 18, ]

# Analyze
summary(lm(score ~ condition, data = data))

# Make plot
ggplot(data, aes(x = condition, y = score)) +
  geom_boxplot()
```

> One line change, rerun, and everything updates.

Why scripting?

Scripting ensures **transparent** and **reproducible** workflows.

Reproducible: You can rerun them.

Documented: You can see what you did and when.

Shareable: Others can inspect and reproduce your analysis.

and RStudio

- **R**: Free, open-source, with thousands of packages for analysis.
- **RStudio**: Intuitive interface for coding, plotting, and debugging.
- Vibrant **community** for support and resources.

Writing better code

- **Name descriptively**: Use `snake_case` or `camelCase` for readability.
- **Comment clearly**: Document your logic for clarity.
- **Organize scripts**: Load packages and data upfront.

Use descriptive names

```r
# Bad
x1 <- c("UNIPD psychology", "university of padova medicine", "unito_biology

# Better
uniDep <- c("unipdPsy", "unipdMed", "unitoBio")
```

Comments, comments and comments…

Write the code for your future self and for others, not for yourself right now.

```r
study_final <- study_raw |>

  # 1. Focus on participants who completed the primary outcome
  filter(!is.na(anxiety_score)) |>

  # 2. Convert raw scores to clinical categories
  mutate(severity = if_else(anxiety_score > 15, "High", "Low")) |>

  # 3. Drop pilot-phase data (pre-2024)
  filter(date >= as.Date("2024-01-01"))
```

Try to open a (not well documented) old code after a couple of years and you will understand :)

Organized scripts

Global operations at the beginning of the script:

- loading packages
- changing general options (`options()`)
- loading datasets

```
# load packages
library(tidyverse) # manipulation
library(ggplot2) # graphic

# options
options(scipen = 999) # digits
set_theme(theme_classic(base_size = 16)) # theme for plots

# load data
dat <- read.csv(...)
```

Functions to avoid repetition

Functions are the primary building blocks of your program. You write small, **reusable**, self-contained **functions** that do one thing well, and then you combine them.

Avoid repeating the same operation multiple times in the script. The rule is, if you are doing the same operation more than two times, write a function.

A function can be re-used, tested and changed just one time affecting the whole project.

Functional programming, example…

We have a dataset (`mtcars`) and we want to calculate the mean, median, standard deviation, minimum and maximum of each column and store the result in a table.

```
1 head(mtcars, n = 3) # display first 3 rows
```

```
                mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4      21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag  21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710     22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
```

```
1 str(mtcars) # show structure
```

```
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
```

Imperative programming

The standard (~imperative) option is using a `for` loop, iterating through columns, calculate the values and store into another data structure.

```
ncols <- ncol(mtcars) # number of columns
# create vectors of length ncols with 0s
means <- medians <- mins <- maxs <- rep(0, ncols)

# loop over the columns (variables) and fill the vectors
for(i in 1:ncols){
  means[i] <- mean(mtcars[[i]])
  medians[i] <- median(mtcars[[i]])
  mins[i] <- min(mtcars[[i]])
  maxs[i] <- max(mtcars[[i]])
}
```

```r
# combine everything into a df
results <- data.frame(means, medians, mins, maxs)
results$col <- names(mtcars) # add variable names

head(results, n = 3) # display 3 rows
```

```
      means medians mins  maxs  col
1  20.09062    19.2 10.4  33.9  mpg
2   6.18750     6.0  4.0   8.0  cyl
3 230.72188   196.3 71.1 472.0 disp
```

Functional programming

The main idea is to decompose the problem writing a function and loop over the columns of the dataframe:

```
summ <- function(x){
  data.frame(means = mean(x), #given an input compute stats
             medians = median(x),
             mins = min(x),
             maxs = max(x))
} # return a df

ncols <- ncol(mtcars) #number of columns
dfs <- vector(mode = "list", length = ncols) #empty list

for(i in 1:ncols){
  dfs[[i]] <- summ(mtcars[[i]])
  #each element of the list is a df with the summary stat
}
```

```
[[1]]
     means medians mins maxs
1 20.09062    19.2 10.4 33.9

[[2]]
   means medians mins maxs
1 6.1875       6    4    8

[[3]]
     means medians mins maxs
1 230.7219   196.3 71.1  472

[[4]]
     means medians mins maxs
```

Functional programming

```r
#combine list to obtain data.frame
results <- do.call(rbind, dfs)
results$var <- names(mtcars) # add variable names
head(results, n = 3) # display 3 rows
```

```
      means medians mins  maxs  var
1  20.09062    19.2 10.4  33.9  mpg
2   6.18750     6.0  4.0   8.0  cyl
3 230.72188   196.3 71.1 472.0 disp
```

Functional programming

```r
ncols <- ncol(USArrests) #number of columns
dfs <- vector(mode = "list", length = ncols)

for(i in 1:ncols){
  dfs[[i]] <- summ(USArrests[[i]])
}

results <- do.call(rbind, dfs)
results$var <- names(USArrests)
head(results, n = 3) # display 3 rows
```

```
    means medians mins  maxs      var
1   7.788    7.25  0.8  17.4   Murder
2 170.760  159.00 45.0 337.0  Assault
3  65.540   66.00 32.0  91.0 UrbanPop
```

Functional programming, `*apply` 📦

- The `*apply` family is one of the best tool in R. The idea is pretty simple: apply a function to each element of a list.
- The powerful side is that in R everything can be considered as a list. A vector is a list of single elements, a dataframe is a list of columns etc.
- Internally, R is still using a `for` loop but the verbose part (preallocation, choosing the iterator, indexing) is encapsulated into the `*apply` function.

```
means <- rep(0, ncol(mtcars))

for(i in 1:length(means)){
  means[i] <- mean(mtcars[[i]])
}

# the same with sapply
means <- sapply(mtcars, mean)
```

The `*apply` family

Apply **your** function…

```
1 results <- lapply(mtcars, summ)
```

Now results is a list of data frames, one per column.

We can stack them into one big data frame:

```
1 results_df <- do.call(rbind, results)
2 head(results_df, n = 5)
```

```
          means medians  mins   maxs
mpg   20.090625  19.200 10.40  33.90
cyl    6.187500   6.000  4.00   8.00
disp 230.721875 196.300 71.10 472.00
hp   146.687500 123.000 52.00 335.00
drat   3.596563   3.695  2.76   4.93
```

Using `sapply`, `vapply`, and `apply`

- `lapply()` always returns a list.
- `sapply()` tries to simplify the result into a vector or matrix.
- `vapply()` is like `sapply()` but safer (you specify the return type).
- `apply()` is for applying functions over rows or columns of a matrix or data frame.

`for` loops are bad?

`for` loops are the core of each operation in R (and in every programming language). For complex operation they are more readable and effective compared to `*apply`. In R we need extra care for writing efficent `for` loops.

Extremely slow, no preallocation:

```
res <- c()
for(i in 1:1000){
  # do something
  res[i] <- i^2
}
```

Very fast:

```
res <- rep(0, 1000)
for(i in 1:length(res)){
  # do something
  res[i] <- i^2
}
```

microbenchmark 📦

```r
library(microbenchmark)

microbenchmark(
  grow_in_loop = {
    res <- c()
    for (i in 1:10000) {
      res[i] <- i^2
    }
  },
  preallocated = {
    res <- rep(0, 10000)
    for (i in 1:length(res)) {
      res[i] <- i^2
    }
  }, times = 100)[1:2,1:2]
```

```
Unit: microseconds
         expr      min       lq     mean   median       uq      max neval
 grow_in_loop 1429.506 1429.506 1429.506 1429.506 1429.506 1429.506     1
 preallocated  960.097  960.097  960.097  960.097  960.097  960.097     1
```

Going further: custom function lists

Let's define a list of functions:

```
funs <- list(mean = mean, sd = sd,
             min = min, max = max, median = median)
```

Now we can apply all of these to every column:

```
# MARGIN = 2 = column
sapply(funs, function(f) apply(mtcars, MARGIN = 2, FUN = f))
```

```
            mean           sd    min     max  median
mpg   20.090625    6.0269481 10.400  33.900  19.200
cyl    6.187500    1.7859216  4.000   8.000   6.000
disp 230.721875 123.9386938 71.100 472.000 196.300
hp   146.687500   68.5628685 52.000 335.000 123.000
drat   3.596563    0.5346787  2.760   4.930   3.695
wt     3.217250    0.9784574  1.513   5.424   3.325
qsec  17.848750    1.7869432 14.500  22.900  17.710
vs     0.437500    0.5040161  0.000   1.000   0.000
am     0.406250    0.4989909  0.000   1.000   0.000
gear   3.687500    0.7378041  3.000   5.000   4.000
carb   2.812500    1.6152000  1.000   8.000   2.000
```

This gives you a matrix with rows as variables and columns as statistics.

Test your functions - `fuzzr` 📦

When you write your own functions, you must test them. In R, we can use `fuzzr` to do **property-based testing**.

Define your function…

```
my_mean <- function(x, na.rm = TRUE
  if (!is.numeric(x)) stop("`x` mus
  if (length(x) == 0) return(NA)
  if (na.rm) x <- x[!is.na(x)]
  if (length(x) == 0) return(NA)
  sum(x) / length(x)
}
```

Define properties that should always hold true…

```
property_mean_correct <- function(x
  x_no_na <- x[!is.na(x)] #remove N
  if (length(x_no_na) == 0) return(
  abs(my_mean(x) - mean(x, na.rm =
}
```

This runs the property on different random numeric vectors and checks whether it holds.

```r
# Property-based testing with 'fuzzr'
library(fuzzr)
test = fuzz_function(fun = property_mean_c
                     arg_name = "x",
                     tests = test_dbl())
lapply(test, function(res) res$test_result
```

```
[[1]]
[1] TRUE

[[2]]
[1] TRUE

[[3]]
[1] TRUE

[[4]]
[1] TRUE

[[5]]
[1] TRUE
```

```r
fuzzr::test_dbl()
```

```
$dbl_empty
numeric(0)

$dbl_single
[1] 1.5

$dbl_mutliple
[1] 1.5 2.5 3.5

$dbl_with_na
[1] 1.5 2.5  NA

$dbl_single_na
[1] NA
```

Why functional programming?

- We can write less and reusable code that can be shared and used in multiple projects.
- The scripts are more compact, easy to modify and less error prone (imagine that you want to improve the `summ` function, you only need to change it once instead of touching the `for` loop).
- Functions can be easily and consistently documented (see roxygen documentation) improving the reproducibility and readability of your code.

Import your functions

You can write some R scripts only with functions and `source()` them into the global environment.

```
project/
│
├─ R/
│   │
│   ├─ utils.R
│   │
│   ├─ analysis.R
```

```r
source("R/utils.R")
results <- lapply(mtcars, summ)
results_df <- do.call(rbind, results)
```

More about functional programming in R

- Advanced R by Hadley Wickham, section on Functional Programming (https://adv-r.hadley.nz/fp.html)
- Hands-On Programming with R by Garrett Grolemund https://rstudio-education.github.io/hopr/
- Hadley Wickham: The Joy of Functional Programming (for Data Science) (https://www.youtube.com/watch?v=bzUmK0Y07ck)

Wrapping up

- Avoid repetition by using functions.
- Test your functions.
- The `*apply` functions are your friends, or `*map` from `purrr` 📦

Organize your project

R Projects

R Projects are a feature implemented in RStudio to organize a working directory.

- They automatically set the working directory
- They allow the use of ***relative paths*** instead of ***absolute paths***
- They provide quick access to a specific project

The working directory problem

How many times have you opened an R script and seen this at the top?

```
1 setwd("Users/tita/sleep-memo/data/raw.csv") #change
```

Instead of hardcoding paths, we want to use projects with **relative paths**.

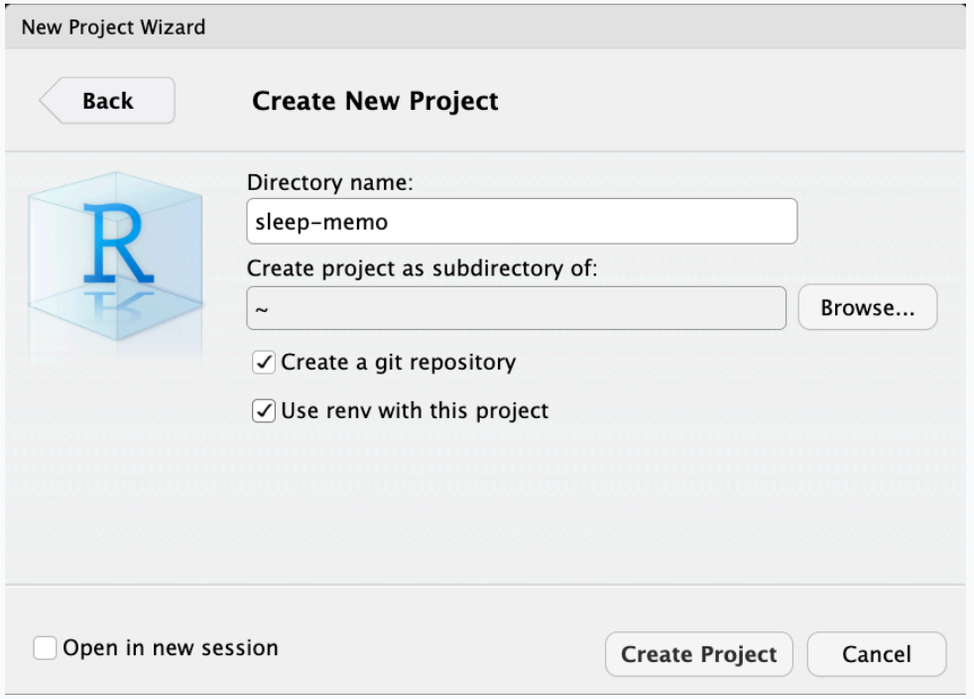
R Projects

An R Project (`.Rproj`) is a file that defines a self-contained workspace.

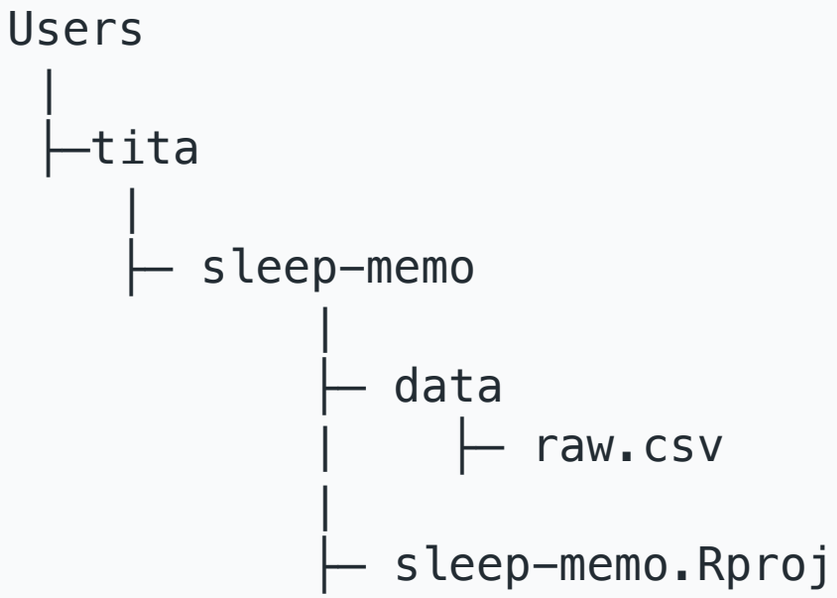
When you open an R Project, your working directory is automatically set to the project root, no need to use `setwd()` ever again.

Open RStudio

`File → New Project → New Directory → New Project`



Relative Path (to the working directory)



Absolute path:

```
1 read.csv("Users/tita/sleep-memo/data/raw.csv")
```

Relative path:

```
1 read.csv("data/raw.csv")
```

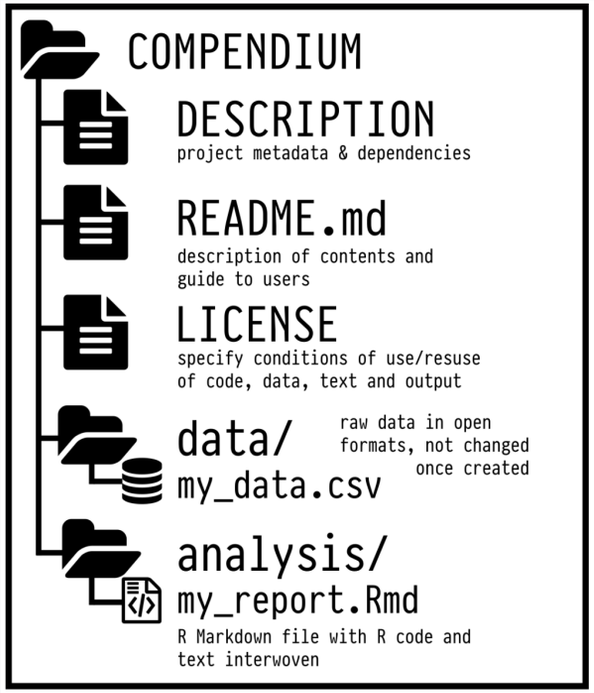
A minimal project structure

```
my-project/
│
├── data/
│   ├── raw/
│   └── processed/
├── R/
│   └── analysis.R
│
├── outputs/
│   ├── figures/
│   └── tables/
│
├── my-project.Rproj
│
└── README.md
```

Project organization with `rrtools` 📦

To make this even "easier", you can use the `rrtools` package to create what's called a reproducible research compendium.

> *… the goal is to provide a standard and easily recognisable way for organising the digital materials of a project to enable others to inspect, reproduce, and extend the research… (Marwick, Boettiger, and Mullen 2018)*



`rrtools::create_compendium("compedium")` builds the basic structure for a research compendium.

> Example

> Tutorial

`renv` 📦 : locking your R environment

Another challenge for reproducibility is package versions.

You write some code today using `dplyr 1.1.2`.

In six months, `dplyr` gets updated… 😢

`renv` helps you create reproducible environments for your R projects!

What does `renv` do?

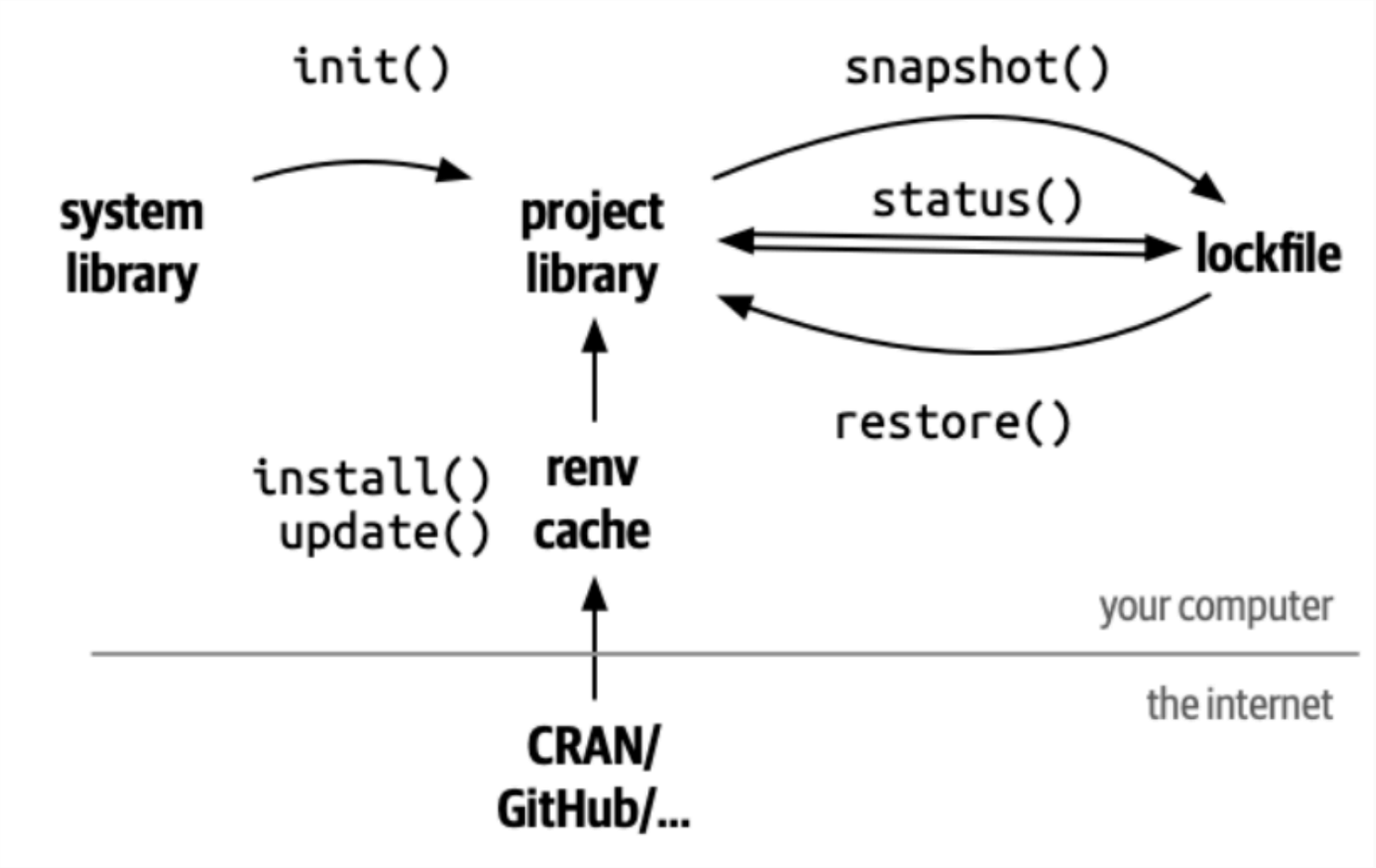
- It records all the packages you use, with versions, in a lockfile
- It installs them in a project-specific library
- It ensures that anyone who runs your code gets exactly the same environment

Project specific library

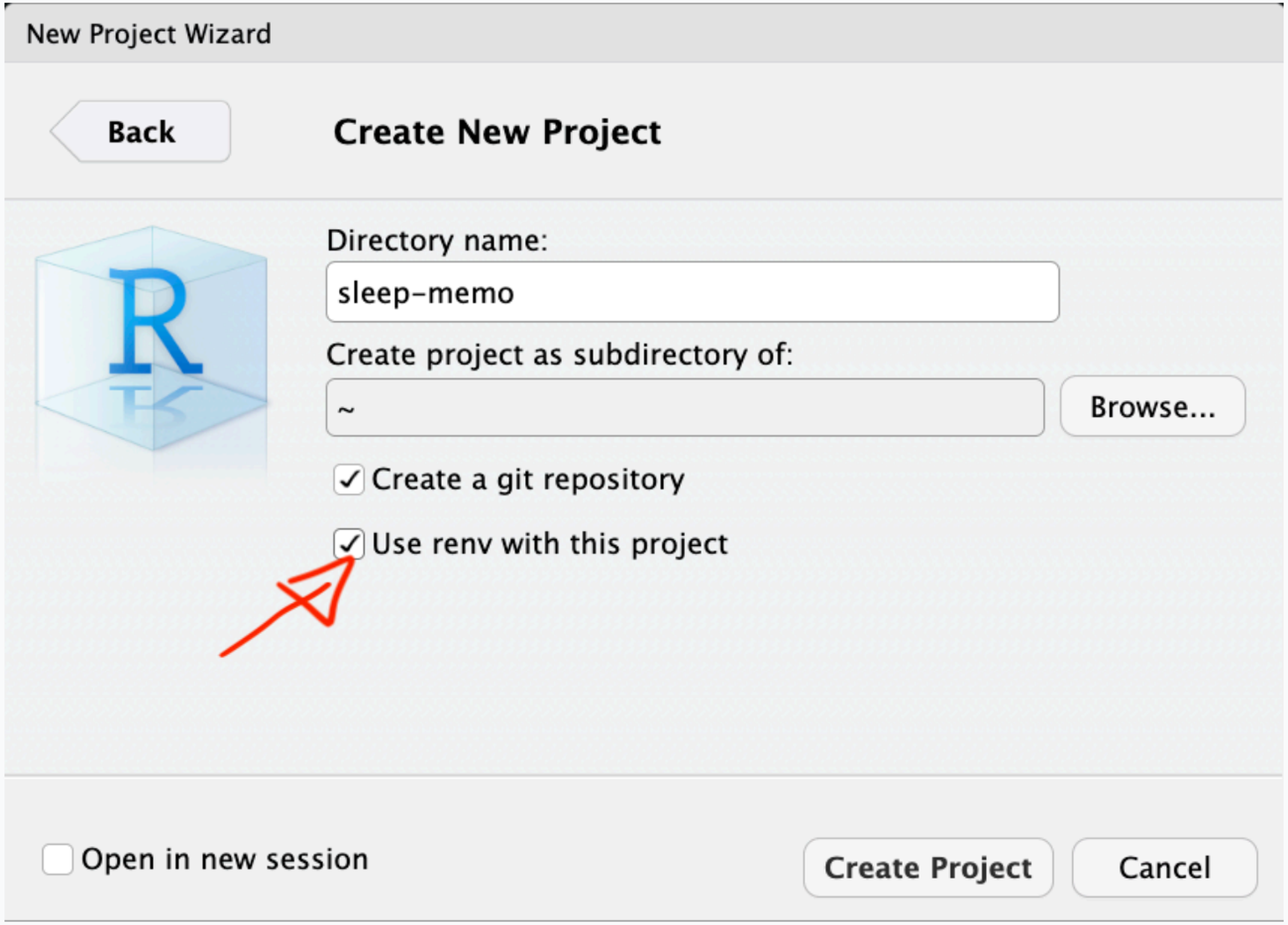
```
install.packages("renv")

renv::init()

install.packages('bayesplot')

# These packages will be installed into
# "~/repro-pre-school/example-renv/renv/library/macos/R-4.4/aarch64-ap# ple
```

> Example

renv commands



`renv::restore()` # re-install from lockfile


New Project Wizard
Back
Create New Project
Directory name:
sleep-memo
Create project as subdirectory of:
~
Browse...
Create a git repository
Use renv with this project
Open in new session
Create Project
Cancel

Research `rrtools` + `renv` 💣

- `rrtools`: Organizes your project into a **reproducible compendium** with clear folders.
- `renv`: Locks **R package versions** for consistent environments.
- Together, they ensure **structure** and **reproducibility** across teams and time.
- Run `rrtools::create_compendium("project")` to start, then `renv::init()` to lock dependencies.

Docker

- Packages your project's **software**, **dependencies**, and **system settings** into a *container*.
- Ensures **consistency** across different computers or servers.
- Ideal for **sharing** complex analyses with others.

Documenting your environment ℹ️

- **`sessionInfo()`**: Captures your **R version**, **packages**, and **platform** in one command.
- Easy way to **document** and **share** your environment.

```
1 sessionInfo()
```

```
R version 4.4.2 (2024-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.6.1

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.0

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Rome
```

Organizing for reproducibility

- Don’t hardcode paths, use `R Projects`
- Create a logical folder structure for your project
- Use `rrtools` to scaffold a research compendium
- Use `renv` to lock your package versions

Literate Programming

What’s wrong about Microsoft Word?

In MS Word (or similar) we need to produce everything outside and then manually put figures and tables.

- writing math formulas
- reporting statistics in the text
- producing tables
- producing plots

Think about the typical MW workflow

- You run your analysis in R
- You copy the results into a Word document
- You tweak the formatting
- You insert a figure generated with R manually
- You change your analysis, but forget to update the results in the text…

Literate Programming

A document where:

- The code is part of the text
- The results are generated ***dynamically***
- The figures are rendered ***automatically***
- Everything is in *sync*

For example **jupyter notebooks**, **R Markdown** and now **Quarto** are literate programming frameworks to integrate code and text.

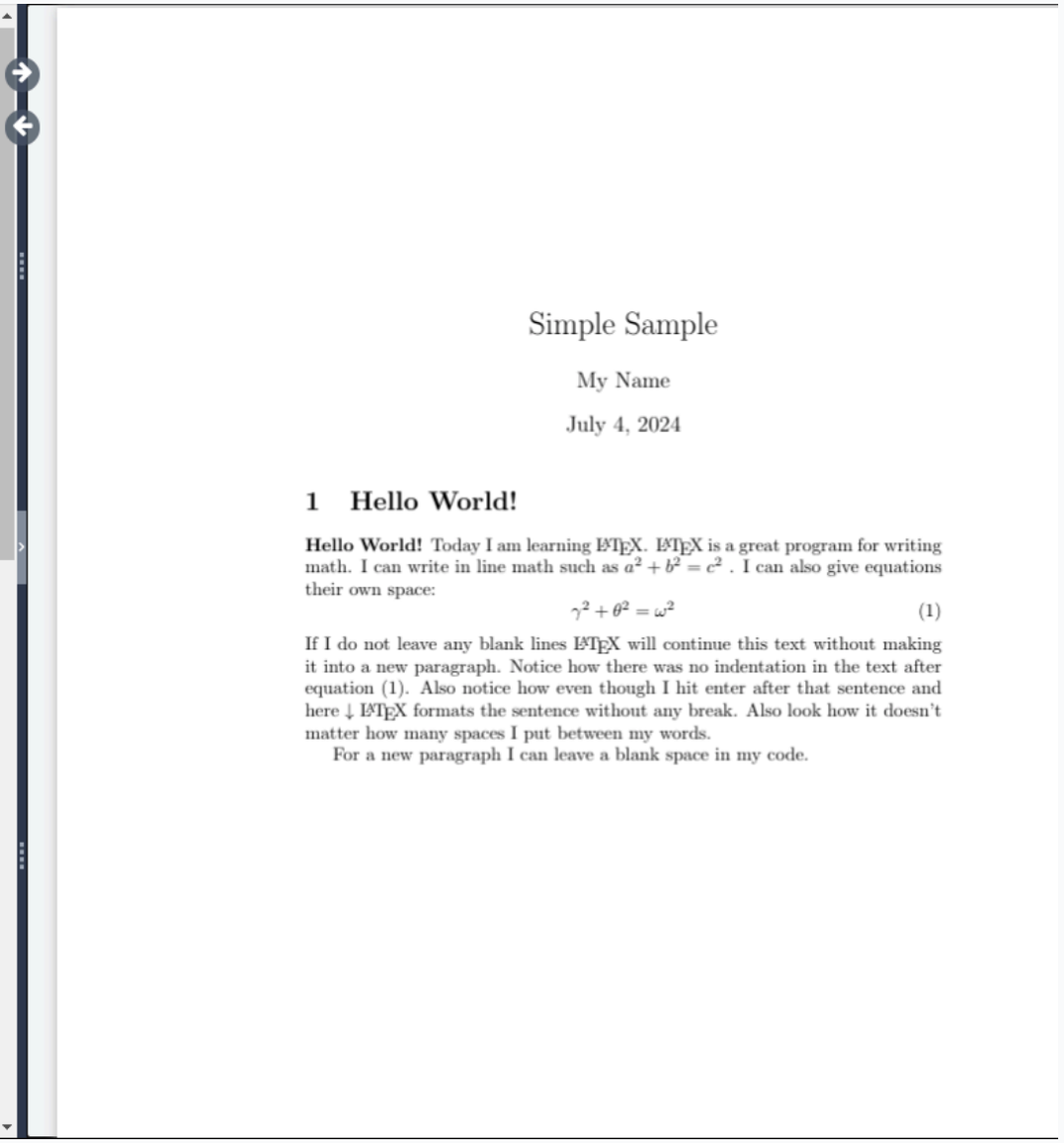
Literate Programming, the markup language

The markup language is the core element of a literate programming framework. When you write in a markup language, you're writing **plain text** while also giving **instructions** for how to generate the final result.

- LaTeX
- HTML
- Markdown
- …

LaTeX

```
% This is a simple sample document.  For more complicated documents take a look in the exercise tab. Note that everything that comes after a % symbol is treated as comment and ignored when the code is compiled.

\documentclass{article} % \documentclass{} is the first command in any LaTeX code.  It is used to define what kind of document you are creating such as an article or a book, and begins the document preamble

\usepackage{amsmath} % \usepackage is a command that allows you to add functionality to your LaTeX code

\title{Simple Sample} % Sets article title
\author{My Name} % Sets authors name
\date{\today} % Sets date for date compiled

% The preamble ends with the command \begin{document}
\begin{document} % All begin commands must be paired with an end command somewhere
    \maketitle % creates title using information in preamble (title, author, date)

    \section{Hello World!} % creates a section

    \textbf{Hello World!} Today I am learning \LaTeX. %notice how the command will end at the first non-alphabet charecter such as the . after \LaTeX
     \LaTeX{} is a great program for writing math. I can write in line math such as $a^2+b^2=c^2$ %$ tells LaTexX to compile as math
     . I can also give equations their own space:
    \begin{equation} % Creates an equation environment and is compiled as math
    \gamma^2+\theta^2=\omega^2
    \end{equation}
    If I do not leave any blank lines \LaTeX{} will continue  this text without making it into a new paragraph.  Notice how there was no indentation in the text after equation (1).
    Also notice how even though I hit enter after that sentence and here $\downarrow$
     \LaTeX{} formats the sentence without any break.  Also  look  how  it  doesn't  matter  how  many  spaces  I put  between  my  words.

```



Simple Sample

My Name

July 4, 2024

1 Hello World!

Hello World! Today I am learning LaTeX. LaTeX is a great program for writing math. I can write in line math such as $a^2 + b^2 = c^2$. I can also give equations their own space:

$$\gamma^2 + \theta^2 = \omega^2 \quad (1)$$

If I do not leave any blank lines LaTeX will continue this text without making it into a new paragraph. Notice how there was no indentation in the text after equation (1). Also notice how even though I hit enter after that sentence and here ↓ LaTeX formats the sentence without any break. Also look how it doesn't matter how many spaces I put between my words.

For a new paragraph I can leave a blank space in my code.

Markdown

Markdown Live Preview Reset Copy Export PDF Sync scroll Dark mode

```
# Markdown syntax guide

## Headers

# This is a Heading h1
## This is a Heading h2
###### This is a Heading h6

## Emphasis

*This text will be italic*
_This will also be italic_

**This text will be bold**
__This will also be bold__

_You **can** combine them_

## Lists

### Unordered
```

Markdown syntax guide

Headers

This is a Heading h1

This is a Heading h2

This is a Heading h6

Emphasis

This text will be italic
This will also be italic

Markdown

Markdown is one of the most popular markup languages for several reasons:

- easy to write and read compared to Latex and HTML
- easy to convert from Markdown to basically every other format using `pandoc`

Quarto

Quarto (https://quarto.org/) is the evolution of R Markdown that integrate a programming language with the Markdown markup language. It is very simple but quite powerful.

Basic Markdown

Markdown can be learned in minutes. You can go to the following link https://quarto.org/docs/authoring/markdown-basics.html and try to understand the syntax.

Quarto

You write your documents in **Markdown**, and **Quarto** turns them into:

- HTML reports
- PDF articles
- Word documents
- Slides
- Website
- Academic manuscripts
- …

Quarto

> Example

- If your data changes, your summary table updates.
- If you update your model, your coefficients update.
- If you change a plot's colors, the new version appear, without having to re-export and re-insert anything.

This eliminates a huge source of human error: **manual updates**.

Outputs

Quarto can generate multiple output formats from the same source file.

- A PDF to send to your colleagues
- A Word document for your co-author who hates PDFs
- An HTML report for your own website

Everything from the same source. No duplication. **Synchronization**.

> Example

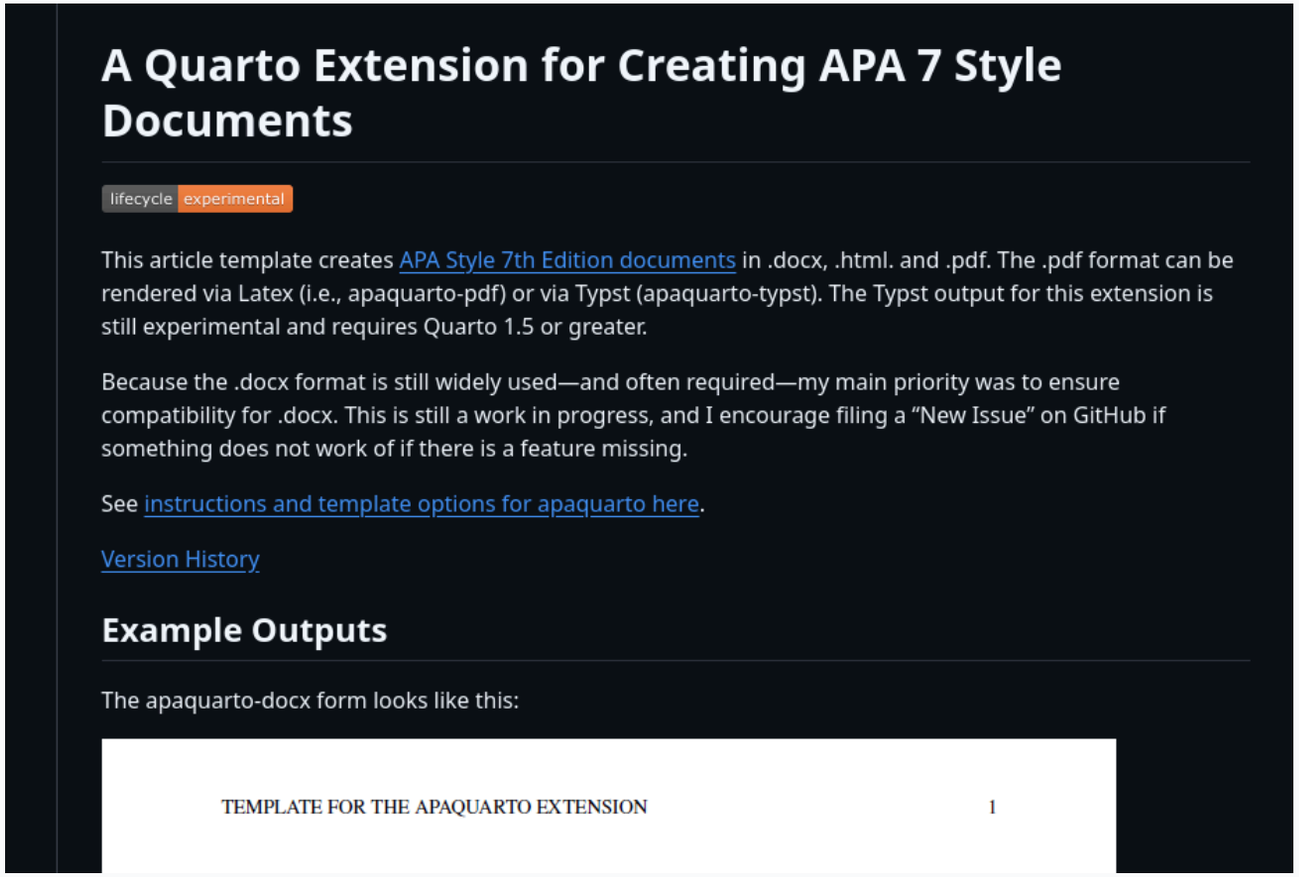
Extra Tools: citations and cross-referencing

- Citations with BibTeX or Zotero
- Cross-references for figures and tables
- Numbered equations with LaTeX syntax
- Footnotes, tables of contents, and more

You can write scientific documents that look and behave just like journal articles, without ever opening Word.

Writing Papers - APA quarto

APA Quarto is a Quarto extension that makes it easy to write documents in APA 7th edition style, with automatic formatting for title pages, headings, citations, references, tables, and figures.



A Quarto Extension for Creating APA 7 Style Documents

lifecycle experimental

This article template creates APA Style 7th Edition documents in .docx, .html. and .pdf. The .pdf format can be rendered via Latex (i.e., apaquarto-pdf) or via Typst (apaquarto-typst). The Typst output for this extension is still experimental and requires Quarto 1.5 or greater.

Because the .docx format is still widely used—and often required—my main priority was to ensure compatibility for .docx. This is still a work in progress, and I encourage filing a "New Issue" on GitHub if something does not work of if there is a feature missing.

See instructions and template options for apaquarto here.

Version History

Example Outputs

The apaquarto-docx form looks like this:

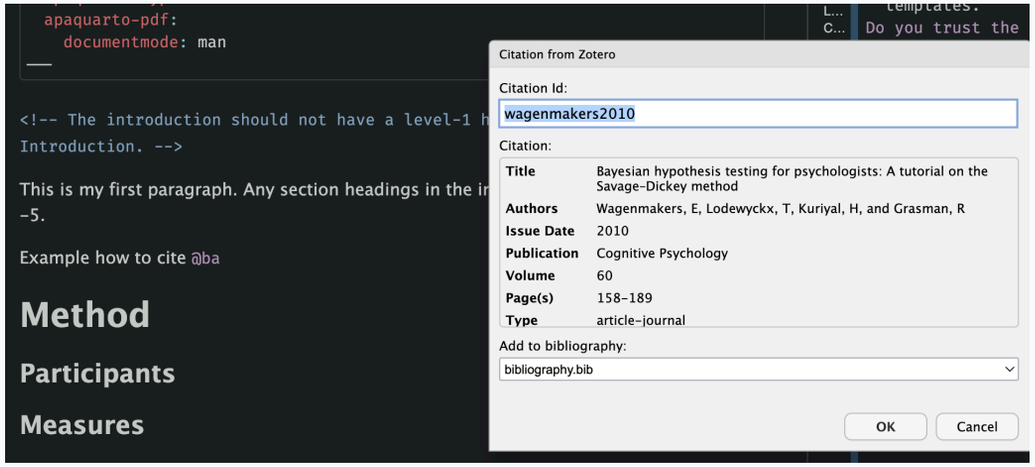
TEMPLATE FOR THE APAQUARTO EXTENSION 1

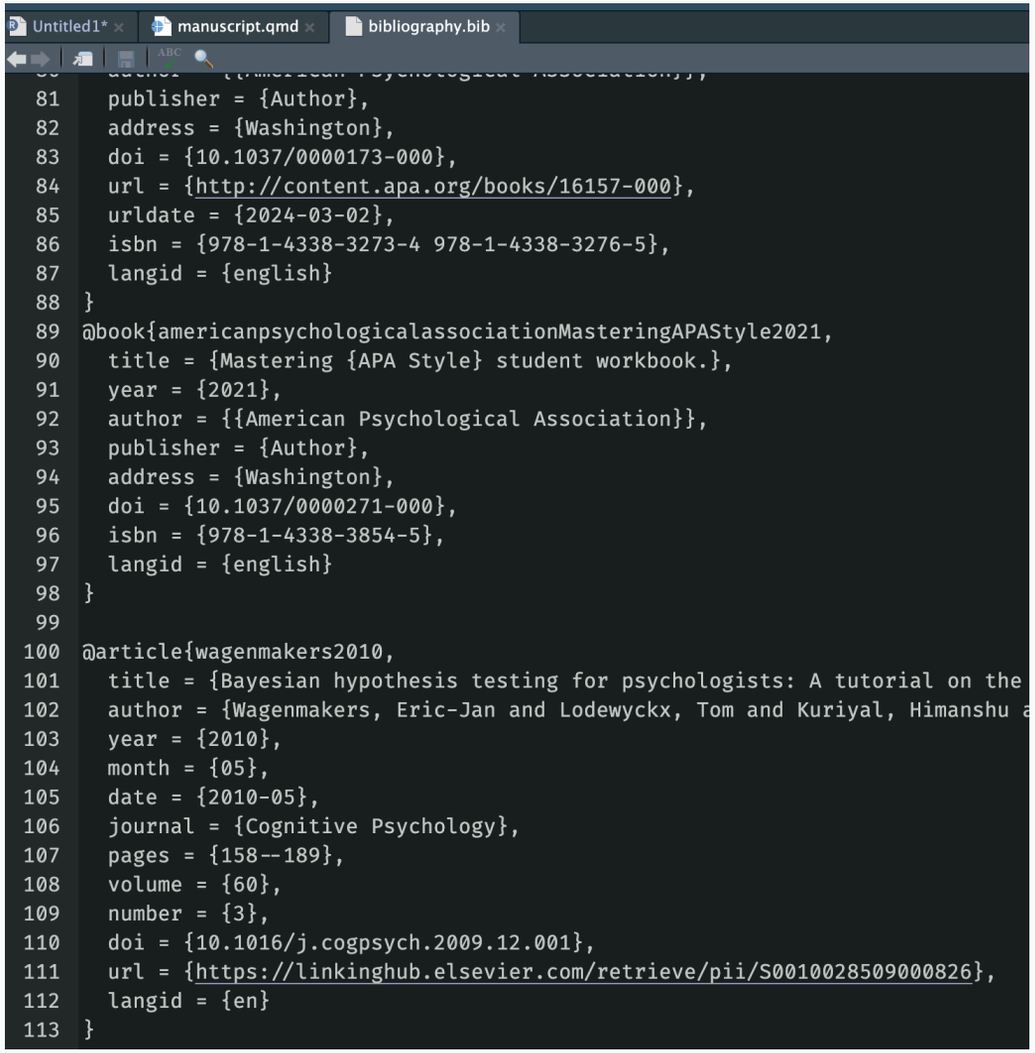
Let’s see an example…

> Example

Quarto + Zotero

Choose your reference:





More about Quarto and R Markdown

The topic is extremely vast. You can do everything in Quarto, a website, thesis, your CV, etc.

- Yihui Xie - R Markdown Cookbook https://bookdown.org/yihui/rmarkdown-cookbook/
- Yihui Xie - R Markdown: The Definitive Guide https://bookdown.org/yihui/rmarkdown/
- Quarto documentation https://quarto.org/docs/guide/

Version Control

Why version control?

You’re working on a project. You save your script as:

- `analysis.R`
- `analysis2.R`
- `analysis_final.R`
- `analysis_final_revised.R`
- `analysis_final_revised_OK_for_real.R`

> No way to know **what changed**, **when**, or **why** — and no safe way to go back.

What is Git?

Git is a **version control system**: it tracks every change you make to your files, so you can always go back to a previous state.

```
git init   # turn any folder into a Git repository
```

You save progress by making **commits**: each one is a labeled snapshot of your project.

```
git add analysis.R
git commit -m "Add initial analysis"
```

> Each commit records: - The changed files - The exact changes - The time - A message describing what you did

Commit message ✍️

A commit message tells your future self (and collaborators) **what you did and why**.

- Write meaningful messages:
- ✅`"Fix bug in anxiety scoring function"`
- ❌`"stuff"`
- Use the imperative mood: `"Add README"`, `"Update plots"`

GitHub

Git tracks your project **on your computer**. GitHub is the **online platform** where you can:

- Back up your project safely in the cloud
- Share it publicly or privately with others
- Collaborate without overwriting each other’s work
- Track issues and project progress

Git workflow

Files move through **three local stages** before reaching GitHub:

> **Remember:**
> New files are **untracked** until you run `git add`.

GitHub in practice

```
git init                                # turn folder into a repo

git add analysis.R                      # stage file for commit

git commit -m "Initial commit"          # save a snapshot locally

git remote add origin <URL>             # link to a GitHub repo

git push -u origin main                 # first push (sets upstream)

git push                                # all subsequent pushes

git pull                                # download others' commits
```

> You can also do most of this from RStudio's Git pane.

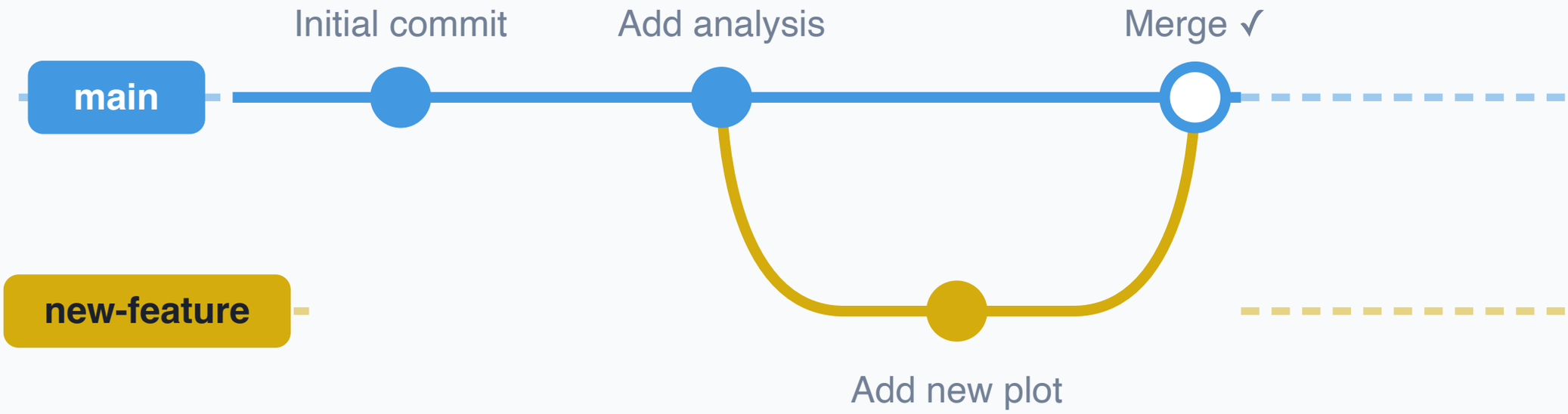
Branching & merging 🌱

By default, you work on the **`main`** branch. A new branch is an **independent copy** of your project where you can experiment safely/without touching the working version.

- Try out new features without breaking `main`
- Fix bugs in isolation
- Let multiple people work in parallel

When the work is ready, you **merge** it back into `main`.

Branching in practice

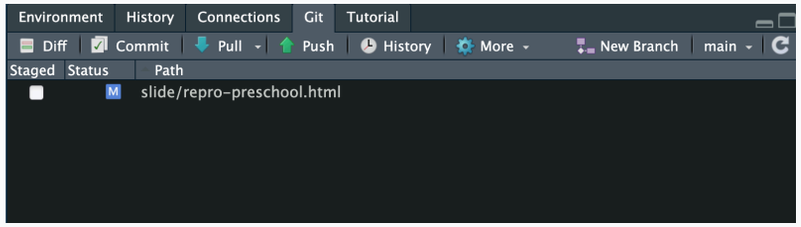



```
git checkout -b new-feature     # 1. Create & switch branch

git add analysis.R              # 2. Commit your changes
git commit -m "Add new plot"

git checkout main               # 3. Switch back to main

git merge new-feature           # 4. Merge branch in
```

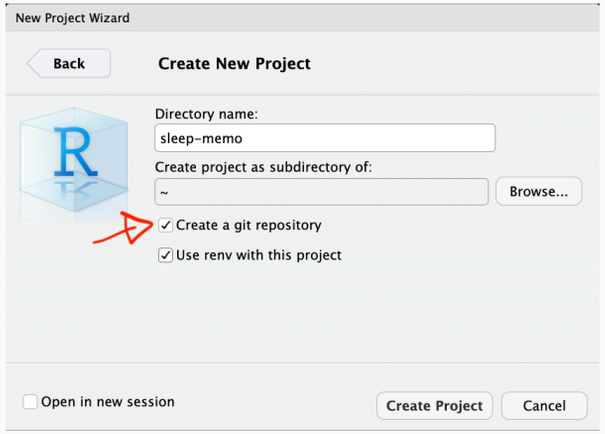
GitHub + RStudio Integration

You don’t have to use the terminal, RStudio has a built-in Git panel, you can:

- **Clone** a repo: `File → New Project → Version Control`
- **Stage, commit, push, pull, browse history**: use the **Git** tab



- **New project with Git**: tick *“Create a git repository”* at setup



Practice & resources

- Happy Git and GitHub for the useR
- GitHub Education: https://education.github.com
- Try GitHub Desktop (GUI client)

If Git and GitHub feel too technical, or if your collaborators are less technical, the OSF is a fantastic alternative or complement.

- Upload data, code, and documents
- Create public or private projects
- Add collaborators
- Create preregistrations
- Generate DOIs for citation
- Track changes

> You can also connect OSF to GitHub.

Integrated workflow 🛠️

1. Develop your analysis using **R and Quarto**.
2. Track code and scripts using **Git**.
3. Host your code on **GitHub** (public or private).
4. Upload your data and materials to **OSF**, including a data dictionary.
5. Link your GitHub repository to your OSF project.
6. Use `renv` for reproducible R environments.
7. Share the OSF project and cite it in your paper.


GO
GREEN
or
GO
HOME!

Reproducibility

It’s about **credibility** and **transparency**.

Reproducible science is **not** about being **perfect**.

It’s about showing your work so that others can **follow**, **understand**, and **build upon** it.

Start simple, don’t wait until you’re “ready”, and teach what you learn!

THANK YOU!

References

Marwick, Ben, Carl Boettiger, and Lincoln Mullen. 2018. “Packaging Data Analytical Work Reproducibly Using r (and Friends).” *The American Statistician* 72 (1): 80–88. https://doi.org/10.1080/00031305.2017.1375986.

Reproducibility, Committee on, Replicability in Science, Board on Behavioral, Cognitive, and Sensory Sciences, Committee on National Statistics, Division of Behavioral, Social Sciences, Education, et al. 2019. *Reproducibility and Replicability in Science*. Washington, D.C.: National Academies Press. https://doi.org/10.17226/25303.

Wilkinson, Mark D., Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, et al. 2016. “The FAIR Guiding Principles for Scientific Data Management and Stewardship.” *Scientific Data* 3 (1): 160018. https://doi.org/10.1038/sdata.2016.18.